Which backend can you actually ship?

On-device ASR runtimes: 4 backends × 3 target platforms. Darker = closer to shippable. Nov 2026

less usable → ship today = does not exist (see footnote for the vendor source)

Hard constraints — CPU-capable · streaming · distributable: only 2 of 12 cells pass.

	x86-64 CPU	x86-64 + NVIDIA GPU	Windows arm64
sherpa-onnx ONNX Runtime	**best** ships today · 96 MB Lite streaming + 5 languages	**ok** CUDA EP exists not enabled / untested here	**ok** win_arm64 wheel exists no hardware to test
sherpa-ncnn ncnn	**ok · runtime untested** SenseVoice ncnn models ⑤ never installed here · no RTF	**ok** Vulkan / GPU path static link, no extra deps	**no wheel** sherpa-ncnn 2.1.15 ships win32 / win_amd64 only ③
GGUF (llama.cpp) ggml	**ok, but non-streaming** SenseVoiceSmall-GGUF 8.01% CER @ 8 threads ④	**ok** llama.cpp CUDA backend single binary, zero deps	**unverified** llama.cpp builds for arm64 no confirmed prebuilt
TensorRT NVIDIA SDK	**does not exist** NVIDIA: no CPU support, GPU or Jetson only ①	**ok, per-device engine** engine bound to build device JetPack unsupported by 11.3 ②	**does not exist** no Windows arm64 target; Jetson is Linux-only ①②

① TensorRT on CPU is not "slow"; it does not exist. NVIDIA's own Triton backend support matrix lists GPU but no CPU for TensorRT, on both x86 and ARM-SBSA.

② TensorRT engines are bound to the device they were built on; hardware compatibility starts at Ampere; JetPack / Jetson is not supported, and 11.3 drops JetPack.

③ Measured via PyPI JSON: sherpa-ncnn 2.1.15 ships cp313 win32 / win_amd64 only, no win_arm64; sherpa-onnx 2.0.1 does ship win_arm64.

④ GGUF ASR is mostly non-streaming today (SenseVoiceSmall-GGUF = SAN-M encoder + CTC, whole-utterance). Its strength is offline batch, not live captions.

⑤ sherpa-ncnn ships prebuilt SenseVoice models (release tag asr-models, 2025-09-13, int8 / float32). What is missing is measurement, not models.

⑥ ncnn has only been run here on an Android device (repacked APK, 2026-10-25), never on Windows x86-64. Android is not a column in this matrix.